

EXP NO :1

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SETTING UP HYPERLEDGER FABRIC DEVELOPMENT ENVIRONMENT

Aim:

To install and configure the Hyperledger Fabric development environment on Linux, create the Fabric test network, and create an application **channel** named "mychannel".

Requirements

- Linux Operating System (Kali)
- Docker and Docker Compose
- Go Programming Language
- Git and Curl
- Hyperledger Fabric Samples, Binaries and Docker Images

Procedure

1. Update the system

Command(s):

- `sudo apt update`
- `sudo apt upgrade -y`

2. Verify Docker installation

Command(s):

- `docker --version`
- `sudo systemctl start docker`
- `sudo systemctl enable docker`
- `docker run hello-world`

```
(mohana@kali)-[~]  
$ docker --version  
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed
```

```
(mohana@kali)-[~]
$ sudo systemctl start docker

(mohana@kali)-[~]
$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
```

```
(mohana@kali)-[~]
$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

3. Verify Go installation

Command(s):

- go version

```
(mohana@kali)-[~]
$ go version
go version go1.26.5 linux/amd64
```

4. Create Fabric workspace

Command(s):

- mkdir -p ~/fabric-dev
- cd ~/fabric-dev

```
(mohana@kali)-[~]
$ mkdir -p ~/fabric-dev

(mohana@kali)-[~]
$ cd ~/fabric-dev/
>

(mohana@kali)-[~/fabric-dev]
$ pwd
/home/mohana/fabric-dev
```

5. Download Hyperledger Fabric

Command(s):

- `curl -sSL https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh -o install-fabric.sh`
- `chmod +x install-fabric.sh`
- `./install-fabric.sh docker samples binary`

```
(mohana@kali)-[~/fabric-dev]
$ curl -sSL https://bit.ly/2ysb0FE | bash -s

Clone hyperledger/fabric-samples repo

=> Changing directory to fabric-samples
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

=> Downloading version 2.5.16 platform specific fabric binaries
=> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.16/hyperledger-fabric-linux-amd64-2.5.16.tar.gz
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
  0     0    0     0    0     0      0      0      0     0  12 121.8M  12 15.34M    0     0  153.7k      0  13:31    0:21    11:49 238.1k^Z
zsh: done      curl -sSL https://bit.ly/2ysb0FE |
zsh: suspended bash -s
```

6. Configure PATH Command(s):

- `echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin'`
`>> ~/.zshrc`
- `source ~/.zshrc`

```
(mohana@kali)-[~/fabric-dev/fabric-samples/test-network]
$ echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.zshrc

(mohana@kali)-[~/fabric-dev/fabric-samples/test-network]
$ source ~/.zshrc
```

7. Verify Fabric binaries

Command(s):

- peer version
- orderer version

```
$ peer version
peer:
Version: v2.5.16
Commit SHA: f871cf9
Go version: go1.26.4
OS/Arch: linux/amd64
Chaincode:
Base Docker Label: org.hyperledger.fabric
Docker Namespace: hyperledger
```

```
(mohana@kali)-[~/fabric-dev]
$ orderer version
orderer:
Version: v2.5.16
Commit SHA: f871cf9
Go version: go1.26.4
OS/Arch: linux/amd64
```

8. Navigate to the test network

Command(s):

- cd ~/fabric-dev/fabric-samples/test-network

9. Start the Fabric network

Command(s):

- ./network.sh up

```

L$ ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb'
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] Running 3/3
✓ Container orderer.example.com Running
✓ Container peer0.org2.example.com Running
✓ Container peer0.org1.example.com Running
CONTAINER ID   IMAGE                                COMMAND                                CREATED        STATUS
342344c842df   hyperledger/fabric-orderer:latest   "orderer"                                About a minute ago Up About a minute
cp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp or
16977a7811dd   hyperledger/fabric-peer:latest     "peer node start"                        About a minute ago Up About a minute
cp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp pe
1000dd744ae1   hyperledger/fabric-peer:latest     "peer node start"                        About a minute ago Up About a minute
cp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp pe
a106a25024e7   hello-world                         "/hello"                                16 minutes ago   Exited (0) 16 minutes ago
518b40179423   hello-world                         "/hello"                                41 hours ago     Exited (0) 41 hours ago

```

10. Verify running containers

Command(s):

- `docker ps`

```

L$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                CREATED        STATUS        PORTS
00de63516518   hyperledger/fabric-orderer:latest   "orderer"                                2 minutes ago Up 2 minutes   0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
19f9b827884f   hyperledger/fabric-peer:latest     "peer node start"                        2 minutes ago Up 2 minutes   0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
43098bbd7945   hyperledger/fabric-peer:latest     "peer node start"                        2 minutes ago Up 2 minutes   0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp

```

11. Create the application channel

Command(s):

- `./network.sh createChannel -c mychannel`

```

$ ./network.sh createChannel -c mychannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] Running 3/3
  ✓ Container orderer.example.com Running
  ✓ Container peer0.org2.example.com Running
  ✓ Container peer0.org1.example.com Running
CONTAINER ID        IMAGE                                COMMAND                  CREATED             STATUS
342344c842df        hyperledger/fabric-orderer:latest  "orderer"               About a minute ago  Up About a minute
cp, [::]:7050→7050/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→9443/tcp, [::]:9443→9443/tcp
16977a7811dd        hyperledger/fabric-peer:latest     "peer node start"       About a minute ago  Up About a minute
cp, [::]:9051→9051/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp
1000dd744ae1        hyperledger/fabric-peer:latest     "peer node start"       About a minute ago  Up About a minute
cp, [::]:7051→7051/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp
a106a25024e7        hello-world                         "/hello"                 17 minutes ago      Exited (0) 17 minutes ago
518b40179423        hello-world                         "/hello"                 41 hours ago        Exited (0) 41 hours ago

Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/mohana/fabric-dev/fabric-samples/test-network/..bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-07-24 16:10:12.342 IST 0001 INFO [common.tools.configtxgen] main → Loading configuration
2026-07-24 16:10:12.350 IST 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization → orderer type
2026-07-24 16:10:12.350 IST 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization → Orderer.Etc
ng to tick_interval:"500ms" election_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapshot_interval_size:16777216
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
  "name": "mychannel",
  "url": "/participation/v1/channels/mychannel",
  "consensusRelation": "consenter",
  "status": "active",
  "height": 1
}

Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-07-24 16:10:18.560 IST 0001 INFO [channelCmd] InitCmdFactory → End
2026-07-24 16:10:18.660 IST 0002 INFO [channelCmd] executeJoin → Success
Joining org1 peer to the channel...

```

12. Stop the network Command(s):

- `./network.sh down`

```
Using docker and docker-compose
Stopping network
[+] Running 7/7
✓ Container orderer.example.com      Removed
✓ Container peer0.org1.example.com    Removed
✓ Container peer0.org2.example.com    Removed
✓ Volume compose_peer0.org1.example.com Removed
✓ Volume compose_orderer.example.com  Removed
✓ Network fabric_test                 Removed
✓ Volume compose_peer0.org2.example.com Removed
```

Result

The Hyperledger Fabric development environment was successfully configured on Linux. The Fabric test network was started successfully, the application channel 'mychannel' was created, and the peer nodes joined the channel successfully.